CampusXI

Product Requirements Document — campus fantasy sports and P2P challenges

Version: 0.3 (draft for review; flat 40% platform fee, ₦400 Mini ticket, 18+ platform-wide)

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Status: Planning. No code written yet. CampusXI is the confirmed product name.

> Inspiration: FantasyFi (play.fantasyfi.io), a Nigerian daily-fantasy football platform. This PRD adopts its core rules (team building, captain multipliers, contest formats, scoring philosophy) and adapts them to university tournament games with manual stat entry, uniform athlete pricing and two P2P betting products. FantasyFi figures quoted here were read from its public rules, FAQ and live contest lobby on 5 November 2026.

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1. Executive summary

1.1 Why now

Inter-faculty and inter-hall tournaments already draw loyal crowds and fierce rivalries on Nigerian campuses. That attention isn't captured anywhere: fixtures go out on WhatsApp, scores get shouted across the pitch, and side bets are informal, unescrowed and argued over afterwards. Meanwhile, daily-fantasy apps such as FantasyFi have taught a generation of students how fantasy contests work, but only for professional leagues.

CampusXI puts that same fantasy layer on top of campus sport. Every school fixture becomes a contest slate, and a fan's knowledge of who actually performs on their campus finally counts.

1.2 What we are building (one sentence)

A mobile-first web app where students build fantasy teams from real campus athletes, enter paid or free contests on each school match, and challenge each other peer-to-peer, with stats entered live by trained student scorers.

1.3 What we are explicitly not building in v1

- Automated data feeds or computer-vision stat capture. All stats are entered manually by scorers.
- Professional-league fantasy. Only campus fixtures are covered.
- Prop bets (single-athlete outcomes), totals, exact-score bets or in-play betting.
- Native iOS/Android apps. v1 is a PWA that installs from the browser.
- Crypto deposits. FantasyFi offers them; CampusXI is NGN only.
- Sports beyond football, basketball and volleyball. Others appear as "coming soon" tiles.

2. Problem statement

2.1 The problem in detail

- **Fans have no stake beyond cheering.** Campus sport has no layer that rewards knowing the athletes, so engagement peaks at finals and fades for group games.
- **Informal betting is already happening, badly.** Students run WhatsApp pools and side bets with no escrow, no clear rules and frequent disputes over who won and who pays.
- **Athletes get no visibility.** There are no stats, no leaderboards and no "player of the tournament" data that a fantasy game produces as a by-product.

2.2 Why this is urgent now, not a someday problem

Fantasy-sports literacy is already high among Nigerian students thanks to national DFS platforms. A campus product that launches during a school sports week inherits a ready-made audience and calendar. Waiting means informal pools become the habit instead.

2.3 Why existing alternatives are insufficient

Alternative	Why it falls short
FantasyFi and other national DFS apps	Professional leagues only. No way to add a campus fixture, athlete or scorer.
WhatsApp/Telegram pools	No escrow, no verified results, no age checks. Disputes are settled by whoever shouts loudest.
Sportsbooks	Don't price campus games. Students bet against the house, not each other.
School sports pages	Fixtures and results only. Nothing interactive.

3. Target customers and personas

3.1 Primary customer segment

Students aged 18+ at a single Nigerian university or polytechnic that runs regular inter-faculty, inter-hall or inter-departmental tournaments in football, basketball and volleyball.

3.2 Secondary segments (post-MVP expansion)

- Alumni and staff of the launch school, who follow rivalries remotely.
- Other universities and polytechnics, onboarded as new "schools" in the same platform.
- Inter-school competitions (e.g. NUGA-style games), where several schools share one slate.

3.3 Primary persona: "Fan-player"

A 19–25-year-old student who follows their faculty team, already uses a DFS or betting app, and has ₦2,000–₦20,000 a month of discretionary money. They want bragging rights over friends, a fair chance to win, and instant withdrawals.

3.4 Persona: "Athlete-player"

A student who plays in the tournament and also wants to play the game. **Athletes may participate fully, including in contests and P2P bets on their own matches** (see the accepted-risk entry in section 13).

3.5 Persona: "Scorer"

A trained student volunteer or sports-council member who records stats from the sideline on a phone. They need a big-button, offline-tolerant entry screen. Scorers may not stake on any match they score.

3.6 Persona: "Sports admin"

A sports-council officer or staff member who sets up tournaments, approves squads, assigns scorers, confirms final results and signs off on settlement.

4. Goals, non-goals, and success metrics

4.1 Product goals for v1 (MVP)

1. Every fixture in a school tournament can be published as a contest slate within 24 hours of the fixture list being released.
2. A scorer can record a full football match with one phone, and live fantasy points update within 10 seconds of each entry.
3. Contests and P2P bets settle automatically and correctly after admin sign-off, with every naira traceable in the ledger.
4. A free mode works end-to-end before any licence is in place, so the product can pilot at a sports week.

4.2 Explicit non-goals for v1

- Covering non-campus sport.
- Revenue from advertising.
- A house-banked sportsbook. CampusXI never takes the other side of a bet.

4.3 Success metrics

Metric	Pilot target (one sports week)	Paid launch target (first semester)
Registered users	1,000	5,000
Users entering ≥1 contest per match day	30% of registered	25% of registered
Average contest fill rate	n/a (free)	≥ 60% of capacity
P2P bet match rate (posted → fully or partly matched)	≥ 40% (0-stake)	≥ 50%
Scorer turnaround (final whistle → stats submitted)	≤ 15 min	≤ 10 min
Result disputes upheld	< 2% of matches	< 1% of matches

5. Functional requirements and user stories

5.1 Epic A — accounts, age and KYC

ID	User story	Acceptance criteria
A1	As a student, I sign up with email or phone and choose a permanent team name.	OTP verification. Team name unique and immutable (FantasyFi rule).
A2	As an 18+ student, I can play free contests without full KYC.	Free mode needs a verified email/phone and a declared date of birth showing 18+. Under-18 sign-ups are refused: the whole platform is 18+.
A3	As a user, I verify identity to unlock paid play.	NIN or BVN check via a KYC provider, plus DOB ≥ 18. Name match. One verified account per identity.
A4	As a user, I set deposit and daily stake limits or self-exclude.	Limits take effect immediately when lowered and after 24h when raised. Self-exclusion blocks all paid actions for the chosen period.

5.2 Epic B — tournaments, teams and athletes

ID	User story	Acceptance criteria
B1	As an admin, I create a tournament with a sport, teams and fixtures.	CSV import of fixtures. Each match has home and away teams, venue, scheduled start and lock time (= scheduled start).
B2	As an admin, I register squads with positions.	Each athlete has a name, team, sport, position and photo (optional). Squads stay editable until lock, as on FantasyFi.
B3	As an admin, I publish the announced line-up before kick-off.	A "Lineup out" badge appears on the match card, and starters and bench are marked.

5.3 Epic C — team builder

ID	User story	Acceptance criteria
C1	As a user, I build a fantasy team for a match.	Every athlete has the **same price**: there is no credit budget. The team must meet the sport's position rules and per-side cap (section 6.1).
C2	As a user, I pick a Captain, Vice-captain and Key player.	C ×2.5, VC ×2, Key ×1.5. All three must be different athletes.
C3	As a user, I create several teams for the same match.	Up to **10 teams per user per match**. A contest may set a lower cap. Duplicate teams (same picks and roles) are rejected.
C4	As a user, I edit my teams until lock.	Unlimited edits until the scheduled start. After that, teams are read-only.

5.4 Epic D — contests and payout engine

ID	User story	Acceptance criteria
D1	As a user, I browse contests for a match and see the prize table before joining.	Each card shows entry fee, spots, prize pool, ranks paid, 1st prize and team cap. A prize-table preview is computed by the payout engine.
D2	As a user, I join a paid contest from my wallet.	Entry fee held on join. Every match has a **~~₦~~400 Mini ticket** contest plus standard contests from ~~₦~~1,000. Maximum 1,000 entrants per contest.
D3	As an admin, I create contests from templates.	Size, fee, rake, minimum fill, guaranteed flag and team cap. The engine refuses configurations it can't fund (section 6.8).
D4	As a user, I'm refunded if a contest doesn't run.	Below minimum fill at lock, or the match is abandoned, means automatic full refund.

5.5 Epic E — scorer app

ID	User story	Acceptance criteria
E1	As a scorer, I record events live with large buttons.	Tap athlete then event (goal, assist, save, card…). Each tap is an append-only event with a timestamp. Undo creates a reversing event.
E2	As a scorer, I keep recording when the signal drops.	Events queue offline and sync in order when the connection returns.
E3	As a second scorer, I record the same match independently.	The system compares both event streams and flags mismatches for the admin.
E4	As a scorer, I submit the final stats and result.	A submission locks the scorer's stream. The admin then reviews it.

5.6 Epic F — scoring engine and live leaderboards

ID	User story	Acceptance criteria
F1	As a user, I watch my points update live.	Points are recomputed from the event stream within 10 seconds and pushed via Supabase Realtime.
F2	As a user, I can see why I have the points I have.	Per-athlete breakdown (e.g. "Goal (MID) +50, Assist +20, ×2.5 Captain").
F3	As a user, I see contest ranks during and after the match.	Leaderboard per contest plus a season-long free tournament leaderboard.

5.7 Epic G — wallet, payments and payouts

ID	User story	Acceptance criteria
G1	As a verified user, I deposit NGN.	Paystack (Flutterwave fallback). Card, transfer and USSD. Deposits credit the ledger only on a verified webhook.

ID	User story	Acceptance criteria
G2	As a user, I withdraw winnings to my bank.	Withdrawals go only to an account in the KYC name. They're paid after the 24h dispute hold on any pending settlement.
G3	As a user, I see every transaction.	Wallet history mirrors the append-only ledger: deposits, holds, releases, wins, refunds, commissions.

5.8 Epic H — P2P Fantasy H2H challenges

ID	User story	Acceptance criteria
H1	As a user, I challenge others with one of my fantasy teams.	Pick a match, a team and a stake (≥ ₦1,000). Post to the open board or to a username or link. The stake is held at once.
H2	As a user, I accept a challenge with my own team.	Exactly one taker per challenge. The creator's team stays hidden until lock. Self or linked accounts are blocked.
H3	As both players, we get settled automatically.	The higher total takes both stakes minus 10% commission on profit. A tie is refunded with no fee.

5.9 Epic I — P2P Match-result bets

ID	User story	Acceptance criteria
I1	As a user, I back Home or Away with a stake.	The card shows the net payout, e.g. "Stake ₦5,000 → win up to ₦9,500".
I2	As a challenger, I take the other side of someone's bet.	The system assigns the opposite side automatically. I choose any amount up to the unmatched balance (≥ ₦1,000) and see my net payout before confirming.
I3	As a bettor, I can be matched by many challengers.	Partial fills from multiple challengers until my stake is fully matched. The unmatched remainder is refunded at lock.
I4	As all parties, we're settled on the confirmed result.	Bettor wins → takes all matched stakes minus commission. Bettor loses → each challenger wins their slice minus commission. Draw → void, everyone refunded.

5.10 Epic J — private leagues, referrals, Predict & Win (post-MVP)

ID	User story	Acceptance criteria
J1	As a user, I create a private contest with a join code.	Equivalent to FantasyFi's "Load Code". Same payout engine.
J2	As a user, I refer friends.	Referral bonus credited as non-withdrawable bonus cash after the friend's first paid entry.
J3	As a user, I play free Predict & Win.	Pick winners for a match day. Free with sponsor prizes.

5.11 Epic K — admin and integrity

ID	User story	Acceptance criteria
K1	As an admin, I confirm the final result and stats.	Compare both scorers' streams, resolve mismatches, confirm. Confirmation starts the 24h dispute hold.
K2	As a user, I can dispute a result within the hold window.	A dispute pauses payout for that match until an admin resolves it. The outcome is logged.
K3	As an admin, I see linked-account and anomaly flags.	Same device, identity or payout account across users, and repeated P2P pairs.
K4	As an admin, I abandon or postpone a match.	Abandoned → all contests and bets refunded. Postponed → lock moves with the new start time, and users may withdraw entries.

6. Scoring and economics specification

6.1 Team composition by sport

Sport	Team size	Position rules	Max from one side	Pricing
Football	11	1 GK · 3–5 DEF · 3–5 MID · 1–3 FWD	7 (FantasyFi rule)	Uniform: no budget
Basketball	8	≥2 Guards · ≥2 Forwards · ≥1 Centre	5	Uniform: no budget
Volleyball	7	≥1 Setter · ≥2 Attackers (OH/OPP) · ≥1 Middle · ≥1 Libero	4	Uniform: no budget
Coming soon	—	Table tennis, tennis, athletics, handball… shown as tiles	—	—

> **Uniform pricing.** Every athlete costs the same, so users don't manage a budget. The trade-off is that more users will pick the same standout athletes, which produces more identical scores. CampusXI handles this with the C/VC/Key multipliers, up to 10 teams per match, and an equal split of prizes between tied ranks (section 6.10). Dynamic pricing (by pick rate or form) is a possible later lever and is not in v1.

6.2 Multipliers (all sports, from FantasyFi)

Role	Multiplier	Rule
Captain (C)	×2.5	Applies to all of the athlete's points, including negatives.
Vice-captain (VC)	×2	Must be a different athlete from C.
Key player (K)	×1.5	Must differ from C and VC.

6.3 Football scoring

Based on FantasyFi's table with the hard-to-capture stats removed, so one sideline scorer can track a match. Values are stored in a versioned `scoring_rule_versions` record, so they can be tuned without a code change.

Event	Points	FantasyFi value	Notes
In starting line-up	+4	+4	
Comes on as substitute	+2	+2	
Goal: forward	+40	+40	
Goal: midfielder	+50	+50	
Goal: defender or goalkeeper	+60	+60	
Assist	+20	+20	Final pass or touch before the goal, scorer's judgement

Event	Points	FantasyFi value	Notes
Save (GK)	+6	+6	
Penalty saved (GK)	+50	+50	
Clean sheet (GK/DEF)	+20	+20 (55+ min)	Must be on the pitch ≥ 60% of scheduled match length (school matches vary)
Goal conceded (GK/DEF on pitch)	−2	−2	
Yellow card	−4	−4	
Red card	−10	−10	
Own goal	−8	−8	
Penalty missed	−20	−20	
Shot on target / chance created / per 5 passes / interception / tackle	—	+6 / +3 / +1 / +4 / 0	**Dropped in v1:** too hard to record reliably by hand

6.4 Basketball scoring (draft; finalised after the pilot)

Event	Points
In starting five	+4
Comes off the bench and plays	+2
Each point scored	+1
Rebound	+1.2
Assist	+1.5
Steal	+3
Block	+3
Turnover	−1

6.5 Volleyball scoring (draft; finalised after the pilot)

Event	Points
In starting six (or starting libero)	+4
Comes on as substitute	+2
Kill (attack point)	+3
Service ace	+4
Block point	+3
Dig	+1
Set assist	+1
Service or attack error	−1

6.6 Edge cases

- **Doesn't play:** an athlete who neither starts nor comes on scores 0 (FantasyFi rule). That includes a C, VC or Key pick. There is no automatic vice-captain promotion.
- **Abandoned match:** all contests and P2P bets on it are void and refunded in full.
- **Postponed match:** the lock moves with the new start. Users get a window to withdraw entries for a refund. If the match isn't replayed within 7 days, it is treated as abandoned.
- **Extra time:** counts for fantasy points. Penalty shoot-outs do not count (except for result bets; see 6.12).
- **Stat corrections:** allowed until admin confirmation. After confirmation, only a dispute can change stats, and changes are new reversing events, never edits.

6.7 Contest formats and rake

Pools scale with the number of entries actually received (as on FantasyFi). A contest runs if it reaches its minimum fill at lock (default: 20% of capacity, and at least 10 entries for contests of 11+). Otherwise it is cancelled and refunded. A **Guaranteed** contest runs at full advertised pool regardless of fill, with a sponsor or the platform covering the shortfall.

Platform fee: CampusXI takes a flat 40% of all entry fees in every contest, whatever its size. The remaining 60% is the prize pool, which is shared among the winning ranks using the rules in section 6.8.

Size tier	Example sizes	Platform fee	Prize pool	Target 1st-prize share	FantasyFi observed fee
1v1	2	40%	60%	Winner takes all	10–20%
Small	5, 11, 20	40%	60%	50% of pool	20–23%
Medium	50, 100, 200	40%	60%	25% of pool	~11% (beginner) to 30%
Large	500, 1,000 (max)	40%	60%	15% of pool	~28–30% (mega)

Entry tiers: **₦1,000, ₦2,000, ₦5,000, ₦10,000**. All rakes, tiers and shares are config values per contest template.

Mini ticket (₦400). Every match automatically gets a second, low-budget ticket: a ₦400 contest for players who can't stretch to ₦1,000. It uses the same team rules, 40% platform fee and payout algorithm as standard contests. The 1.2× floor means every Mini profit rank wins at least ₦480 before rounding. Mini contests are still paid play, so 18+ KYC applies. P2P stakes stay at a ₦1,000 minimum (confirmed by the product owner).

> **Confirmed by the product owner:** a flat **40% platform fee** on every contest; the top **40% of contest ranks** are paid (top 25% profit, the rest of the top 40% get their entry back); standard minimum ₦1,000, Mini ticket ₦400; P2P bets keep a 10% commission on winnings with a ₦1,000 minimum stake. **Side effect to watch:** in a 1v1 contest the winner takes only 1.2× their entry, e.g. ₦1,200 back for ₦1,000. Fantasy H2H challenges (10% commission) are the better-value 1v1 option.

6.8 Payout algorithm

The system computes the prize table automatically for any number of entries N, using the actual entries at lock:

1. **Bands.** Paid ranks P = floor(40% × N). Profit ranks W = floor(25% × N), with at least 1 profit rank when N ≥ 3. A 1v1 contest is winner-takes-all.
2. **Break-even band.** Ranks W+1 to P each receive exactly 1× the entry fee: their money back.
3. **Profit pool.** R = prize pool − (P − W) × entry. Every profit rank first gets a floor of 1.2× entry, so a profit rank always profits.
4. **Top-weighting.** The rest of R is shared by weight 1/rank^α. The engine solves α so that 1st prize ≈ the tier's target share of the pool. Prizes never increase as rank goes down.
5. **Rounding.** Prizes are rounded down to ₦10, and any rounding residue goes to 1st. The table always sums to the pool exactly.
6. **Validation.** If the pool can't fund the break-even band plus the profit floors, the contest template is rejected at creation.

6.9 Worked payout examples (generated by the reference engine)

Entries × fee	Fee	Pool	Ranks paid	Ranks in profit	1st prize	Lowest profit prize
2 × ₦1,000	40%	₦1,200	1	1	₦1,200	₦1,200
5 × ₦1,000	40%	₦3,000	2	1	₦2,000	₦2,000
11 × ₦1,000	40%	₦6,600	4	2	₦3,300	₦1,300
100 × ₦1,000	40%	₦60,000	40	25	₦15,040	₦1,200
400 × ₦1,000	40%	₦240,000	160	100	₦36,540	₦1,200
1,000 × ₦1,000	40%	₦600,000	400	250	₦91,110	₦1,200
100 × ₦400 Mini	40%	₦24,000	40	25	₦6,030	₦480
1,000 × ₦400 Mini	40%	₦240,000	400	250	₦36,800	₦480

The 5-entry and 11-entry rows reproduce FantasyFi's own small-contest shapes (5 spots → 2 ranks paid; 11 spots → 4 ranks paid, "win up to 40% rank").

Small: 11 entries × ₦1,000 = ₦11,000 → 40% platform fee → pool ₦6,600

Rank(s)	Prize each (₦)
1	3,300
2	1,300
3–4	1,000 (money back)

Medium: 100 entries × ₦1,000 = ₦100,000 → 40% platform fee → pool ₦60,000

Rank(s)	Prize each (₦)

Rank(s)	Prize each (₦)
1	15,040
2	2,100
3	1,380
4	1,250
5	1,220
6	1,210
7	1,200
8	1,200
9	1,200
10	1,200
11–25	1,200–1,200
26–40	1,000 (money back)

Large (maximum size): 1,000 entries × ₦1,000 = ₦1,000,000 → 40% platform fee → pool ₦600,000

Rank(s)	Prize each (₦)
1	91,110
2	24,180
3	11,620
4	7,150
5	5,050
6	3,890
7	3,190
8	2,740
9	2,420
10	2,190
11–250	1,200–2,020
251–400	1,000 (money back)

Mini ticket (maximum size): 1,000 entries × ₦400 = ₦400,000 → 40% platform fee → pool ₦240,000

Rank(s)	Prize each (₦)
1	36,800
2	9,670
3	4,650
4	2,860

Rank(s)	Prize each (₦)
5	2,020
6	1,550
7	1,270
8	1,090
9	960
10	870
11–250	480–810
251–400	400 (money back)

A 1,000-capacity contest that only fills to 400 entries pays on N = 400: pool ₦240,000, 160 ranks paid, 1st prize ₦36,540.

6.10 Ties

Tied entries share the combined prizes of the ranks they occupy, equally. Example (11-entry contest, prizes ₦3,300 / ₦1,300 / ₦1,000 / ₦1,000): if two entries tie for 2nd, they share ranks 2 and 3, so each gets ₦1,150, and the next entry is ranked 4th and gets ₦1,000. Uniform pricing makes ties more likely, so this rule is part of the payout engine's test suite.

6.11 P2P Fantasy H2H challenges

- **Create:** choose a match, one of your fantasy teams for it, and a stake (minimum ₦1,000; tiers and maximum are configurable). Post to the open board (visible to all users for that match) or send to a username or a shareable link. Your stake is held immediately.
- **Board:** filter by sport, match and stake. Each card shows the creator's display name and H2H record, never their team until lock (so it can't be copied).
- **Accept:** exactly one taker attaches one of their own teams for the match. Their stake is held. Self-acceptance and linked accounts (same device, KYC identity or payout account) are blocked.
- **Edit:** both players can change their teams until lock.
- **Settle:** the higher fantasy total wins both stakes minus a 10% commission on profit. At ₦1,000 each, the winner gets ₦1,900 and CampusXI keeps ₦100. A tie refunds both stakes with no fee.
- **Refunds:** the creator may cancel before acceptance. A challenge unaccepted at lock is refunded. An abandoned match refunds both.

6.12 P2P Match-result bets

- **Create:** choose a match, a side (**Home** or **Away**) and a stake. The stake is the most you can win from challengers. The card shows the net payout: "Stake ₦5,000 → win up to ₦9,500" at 10% commission.
- **Challenge:** a challenger opens the bet and the system assigns them **the opposite side automatically**. They enter any amount from ₦1,000 up to the unmatched balance, and see their net payout (e.g. "Stake ₦1,000 → win ₦1,900") before confirming.

- **Multiple challengers:** a bet stays on the board, partially matched, until challenger slices add up to the full stake. The total matched can **never exceed** the bettor's stake, so every naira of liability is escrowed and no one can lose more than they put up.
- **Bettor's side wins:** the bettor takes **all** matched challenger stakes, minus 10% commission on that profit, plus their own stake back.
- **Opposite side wins:** each challenger gets their slice back plus an equal amount from the bettor's stake, minus 10% commission. The bettor gets back any unmatched remainder.
- **Draw (football):** Home/Away bets are void on a draw ("draw no bet"). Everyone is refunded and no fee is taken. Knockout matches settle on the official winner including penalty shoot-out. Basketball and volleyball have no draws.
- **Lock:** at the scheduled start. The unmatched remainder is refunded at lock. No in-play betting in v1.

Worked example: the bettor stakes ₦5,000 on Home. Three challengers each take ₦1,000 on Away (₦3,000 matched, ₦2,000 unmatched).

Outcome	Bettor receives	Each challenger receives	CampusXI commission
Home wins	₦7,700	₦0	₦300
Away wins	₦2,000 (unmatched refund)	₦1,900	₦300
Draw	₦5,000 (void)	₦1,000 (void)	₦0

> **Licensing.** A result bet is a wager on a sporting outcome, which is sports betting rather than fantasy. Match-result bets are built behind a feature flag and stay **off** until a sports-betting licence is held. Fantasy contests and H2H challenges run under the fantasy licence. Counsel to confirm both (section 11).

6.13 Explainability requirement

- Every fantasy point shown links to the scorer event that produced it.
- Every contest shows its full prize table before entry and freezes it at lock.
- Every P2P bet shows the net payout for each outcome before confirmation.
- Every wallet movement maps to one ledger entry with a reason and reference.

6.14 How CampusXI compares with FantasyFi

Area	FantasyFi	CampusXI
Fixtures	Professional leagues via data feed	Campus tournaments, manual scorers
Team	11 players, credit budget, max 7 per side	11/8/7 by sport, uniform pricing, per-side cap
Multipliers	C ×2.5, VC ×2, Key ×1.5	Same
Teams per match	Multiple (contest caps of 1–12 seen)	Up to 10
Entry fees	From ~₦135	₦400 Mini ticket on every match; standard from ₦1,000
Platform fee	~10–30%	Flat 40% of entry fees, every size
Ranks paid	25–50%	Top 40% paid, top 25% profit

Area	FantasyFi	CampusXI
P2P	Head-to-head contests	H2H challenges + Home/Away bets with multi-challenger fills
Payments	NGN/GHS/KES + crypto	NGN only

7. Data model

7.1 Core entities: relationship summary

- A **school** has many **tournaments**. A tournament has one **sport**, many **teams** and many **matches**.
- A team has many **athletes**. A match has **match_lineups**, a stream of **stat_events** and one confirmed **match_result**.
- A **user** has many **fantasy_teams** (≤ 10 per match), each with **fantasy_team_picks**.
- A match has many **contests**. A contest has a frozen **payout_table** and many **contest_entries** (each linking one fantasy team).
- A match has many **p2p_bets**. Each bet has one or more **p2p_bet_fills** and an append-only **p2p_bet_events** log.
- All money moves through the append-only, double-entry **wallet_ledger**.

> **Money rule:** all amounts are stored as integer **kobo** (`bigint`), never floats. **Append-only tables:** `stat_events`, `p2p_bet_events`, `wallet_ledger`, `audit_log`. Application code never UPDATEs or DELETEs rows in them. Corrections are new reversing rows.

7.2 Schema — users

Accounts for all roles.

Column	Type	Notes
id	uuid PK	Supabase auth user id
school_id	uuid FK	Home school
team_name	text unique	Immutable after creation
roles	text[]	player, scorer, admin, superadmin
kyc_status	enum	none, pending, verified, rejected
dob	date	From KYC; paid play requires age ≥ 18
device_fingerprint	text	Used for linked-account detection
self_excluded_until	timestamptz null	
created_at	timestamptz	

7.3 Schema — athletes

Real campus athletes. Pricing is uniform, so there is no price column in v1.

Column	Type	Notes
id	uuid PK	
team_id	uuid FK	
sport_id	uuid FK	

Column	Type	Notes
display_name	text	
position	text	Sport-specific code (GK, DEF, G, F, C, S, OH, MB, L…)
user_id	uuid FK null	Set if the athlete also has a player account (full participation allowed)
active	bool	

7.4 Schema — matches and match_results

Fixtures and the admin-confirmed outcome.

Column	Type	Notes
matches.id	uuid PK	
matches.tournament_id	uuid FK	
matches.home_team_id / away_team_id	uuid FK	
matches.scheduled_start	timestamptz	Lock time for teams, contests and bets
matches.status	enum	scheduled, lineup_out, live, final_pending, confirmed, abandoned, postponed
match_results.match_id	uuid PK/FK	
match_results.home_score / away_score	int	
match_results.winner	enum	home, away, draw (shoot-out winner recorded separately)
match_results.confirmed_by / confirmed_at	uuid / timestamptz	Starts the 24h dispute hold

7.5 Schema — stat_events (append-only)

One row per scorer tap. Points are always recomputed from this stream.

Column	Type	Notes
id	bigserial PK	
match_id	uuid FK	
scorer_id	uuid FK	Which of the two scorers
athlete_id	uuid FK	
event_type	text	Key in the scoring rule version (goal, assist, save…)
minute	int null	
reverses_event_id	bigint null	Set for undo or correction events
client_seq	int	Offline ordering from the scorer device

Column	Type	Notes
created_at	timestamptz	

7.6 Schema — fantasy_teams and fantasy_team_picks

User teams for one match.

Column	Type	Notes
fantasy_teams.id	uuid PK	
fantasy_teams.user_id / match_id	uuid FK	≤ 10 per (user, match)
fantasy_teams.label	text	"Team 1"…"Team 10"
fantasy_teams.locked_at	timestamptz null	Set at match lock
fantasy_team_picks.fantasy_team_id	uuid FK	
fantasy_team_picks.athlete_id	uuid FK	
fantasy_team_picks.role	enum	none, captain, vice_captain, key

7.7 Schema — contests, payout_tables, contest_entries

Paid and free contests per match.

Column	Type	Notes
contests.id	uuid PK	
contests.match_id	uuid FK	
contests.capacity	int	2 to 1,000
contests.entry_fee_kobo	bigint	0 free; 40,000 (₦400) Mini; ≥ 100,000 (₦1,000) standard
contests.is_mini	bool	Auto-created Mini ticket contest, one per match
contests.rake_bps	int	Platform fee in basis points; default 4000 = 40% for every size
contests.min_fill	int	
contests.guaranteed	bool	
contests.team_cap	int	≤ 10
contests.first_share_bps	int	Target 1st-prize share
contests.status	enum	open, locked, cancelled, settled
payout_tables.contest_id	uuid FK	Frozen at lock for the actual N
payout_tables.rank_from / rank_to / prize_kobo	int / int / bigint	
contest_entries.contest_id / fantasy_team_id	uuid FK	Unique pair

Column	Type	Notes
contest_entries.final_points / final_rank / payout_kobo	numeric / int / bigint	

7.8 Schema — p2p_bets

One table for both P2P kinds.

Column	Type	Notes
id	uuid PK	
kind	enum	fantasy_h2h, match_result
match_id	uuid FK	
creator_id	uuid FK	
creator_side	enum null	home or away (match_result only)
creator_fantasy_team_id	uuid FK null	fantasy_h2h only
stake_kobo	bigint	≥ ₦1,000
matched_kobo	bigint	Sum of fills; constraint matched ≤ stake
visibility	enum	board, direct
invite_token	text unique null	For link or username challenges
commission_bps	int	Default 1000 = 10%
status	enum	open, partially_matched, matched, locked, settled, void, refunded, cancelled

7.9 Schema — p2p_bet_fills

One row per challenger.

Column	Type	Notes
id	uuid PK	
bet_id	uuid FK	
challenger_id	uuid FK	≠ creator, not linked
side	enum null	Auto-set to the opposite of creator_side
challenger_fantasy_team_id	uuid FK null	fantasy_h2h only; exactly one fill
amount_kobo	bigint	fantasy_h2h: = stake; match_result: ≥ ₦1,000 and ≤ unmatched
payout_kobo / commission_kobo	bigint	Set at settlement

7.10 Schema — p2p_bet_events (append-only)

Status history for every bet, mirroring the append-only audit pattern.

Column	Type	Notes
id	bigserial PK	
bet_id	uuid FK	
from_status / to_status	enum	
actor	text	user id or system job name
reason	text	
created_at	timestamptz	

7.11 Schema — wallet_ledger (append-only, double-entry)

Every naira movement as balanced debit/credit lines.

Column	Type	Notes
id	bigserial PK	
txn_id	uuid	Groups the balanced lines of one transaction
account	text	user:{id}:available, user:{id}:held, platform:rake, platform:commission, psp:clearing…
amount_kobo	bigint	Positive = credit, negative = debit. Each txn_id sums to 0
entry_type	enum	deposit, withdrawal, hold, release, entry_fee, prize, stake, payout, refund, rake, commission, bonus
ref_type / ref_id	text / uuid	contest_entry, p2p_bet_fill, deposit…
created_at	timestamptz	

7.12 Supporting tables

Table	Purpose
schools, sports, tournaments, teams	Tenancy and competition structure
scoring_rule_versions	Versioned JSON scoring tables per sport. A match pins the version active at lock
match_lineups	Starters and bench per match
stat_submissions	Each scorer's final submission and the admin's reconciliation
kyc_records	Provider reference, verified name, DOB, status (PII encrypted at rest)
deposits, withdrawals	PSP references, status and webhook payloads
referrals, exclusions	Referral links/bonuses; self-exclusion and limit history
audit_log (append-only)	Admin actions: confirmations, disputes, overrides

8. API contract specification

FastAPI, JSON over HTTPS, bearer JWT from Supabase Auth. All money fields are integer kobo. Errors use `{ "error": { "code", "message" } }`.

8.1 Authentication and KYC

Method	Path	Purpose / rules
POST	/auth/otp	Send OTP to email or phone
POST	/auth/verify	Exchange OTP for session
POST	/me/kyc	Submit NIN/BVN + DOB, returns pending
GET	/me	Profile, roles, KYC status, limits
PUT	/me/limits	Deposit or stake limits; self-exclusion

8.2 Lobby and matches

Method	Path	Purpose / rules
GET	/sports	Live sports plus "coming soon" tiles
GET	/matches?sport=&status=	Upcoming and live matches with lock countdown
GET	/matches/{id}	Squads, line-up, status, result
GET	/matches/{id}/live	Live points snapshot (Realtime channel for updates)

8.3 Team builder

Method	Path	Purpose / rules
POST	/matches/{id}/teams	Create team (picks + C/VC/K). Validates positions, side cap, ≤ 10 teams
PUT	/teams/{id}	Edit until lock
GET	/me/teams?match=	My teams with live points
GET	/teams/{id}/breakdown	Per-athlete point explanation

8.4 Contests

Method	Path	Purpose / rules
GET	/matches/{id}/contests	Contest cards
GET	/contests/{id}/payout-preview?n=	Prize table for N entries (defaults to capacity)
POST	/contests/{id}/entries	Join with a fantasy team. Holds entry fee
DELETE	/entries/{id}	Withdraw before lock (refund)

Method	Path	Purpose / rules
GET	/contests/{id}/leaderboard	Ranks and points

8.5 P2P bets

Method	Path	Purpose / rules
POST	/bets	Create. Body: kind, match_id, side | fantasy_team_id, stake_kobo, visibility, invitee?
GET	/matches/{id}/bets	Open board, filterable by kind and stake
GET	/bets/invite/{token}	Resolve a direct challenge link
GET	/bets/{id}/quote?amount=	Net payout per outcome for a proposed fill
POST	/bets/{id}/fills	Challenge. Side auto-assigned; amount ≤ unmatched; H2H requires fantasy_team_id
POST	/bets/{id}/cancel	Creator cancels the unmatched remainder before lock
GET	/me/bets	My bets and fills

Example: `POST /bets/{id}/fills` with `{ "amount_kobo": 100000 }` returns `{ "fill_id", "side": "away", "amount_kobo": 100000, "win_payout_kobo": 190000, "bet_status": "partially_matched", "unmatched_kobo": 200000 }`.

8.6 Wallet

Method	Path	Purpose / rules
POST	/wallet/deposits	Start a Paystack transaction
POST	/webhooks/paystack	Verified webhook credits the ledger
POST	/wallet/withdrawals	To KYC-name bank account only
GET	/wallet	Available, held, bonus balances + history

8.7 Scorer

Method	Path	Purpose / rules
GET	/scorer/matches	Matches assigned to me
POST	/scorer/matches/{id}/events	Batch of events with client_seq (offline sync)
POST	/scorer/matches/{id}/submit	Final submission

8.8 Admin

Method	Path	Purpose / rules
POST	/admin/tournaments	Create tournament, import fixtures/squads

Method	Path	Purpose / rules
POST	/admin/matches/{id}/lineup	Publish line-up
GET	/admin/matches/{id}/reconcile	Side-by-side scorer streams and mismatches
POST	/admin/matches/{id}/confirm	Confirm result and stats; start dispute hold
POST	/admin/matches/{id}/abandon	Void everything on the match
GET	/admin/flags	Linked accounts and anomalies

8.9 Internal jobs (APScheduler)

Method	Path	Purpose / rules
JOB	lock_match	At scheduled start: lock teams, cancel under-filled contests, freeze payout tables, refund unmatched bet remainders
JOB	settle_match	After hold expires with no open dispute: compute ranks, pay contests and P2P, post ledger entries
JOB	void_draws	On a confirmed draw: void match_result bets and refund
JOB	postponed_sweep	Matches postponed for over 7 days are treated as abandoned

9. Technical architecture and stack

9.1 Guiding principles

- Money correctness over speed: integer kobo, double-entry ledger, idempotent settlement jobs.
- Recompute, don't mutate: points and ranks always derive from the append-only event stream.
- Mobile first: most users are on Android phones with patchy data.
- Choose each layer for this product: Postgres for a ledger that must never drift, Supabase Realtime for live points on fans' phones, Python for engines that are easiest to test exhaustively.

9.2 Recommended stack

Layer	Choice	Why
Frontend	React + TypeScript + Vite, PWA	Installable from the browser, one codebase
Backend	FastAPI (Python)	Scoring and payout engines are pure-Python and easy to test
Database / auth	Supabase Postgres + Auth	Row-level security, managed backups, append-only triggers for the ledger
Data access	psycopg 3 with plain SQL; SQL migrations in supabase/migrations/	Money code reads exactly as it runs; migrations apply with the Supabase CLI
Live updates	Supabase Realtime	Pushes point and leaderboard changes
Jobs	APScheduler	Lock, settle, void and sweep
Email	Resend	OTP fallback, receipts
Payments	Paystack (Flutterwave fallback)	NGN cards, transfer and USSD. **Risk:** both may refuse gaming merchants without a licence
KYC	Nigerian KYC provider (e.g. Prembly or Dojah)	NIN/BVN and DOB

9.3 Live points pipeline

Scorer app → `POST /scorer/.../events` → insert `stat_events` → recompute the affected athletes' points for the pinned scoring version → update cached team totals → broadcast on the match's Realtime channel. Target: under 10 seconds from tap to fan screen.

9.4 Settlement pipeline

Confirm → 24h dispute hold → `settle_match` (idempotent on match id) → one ledger transaction per entry or fill → bet and contest status events → withdrawals unlocked. A rerun produces no new ledger rows.

9.5 Suggested repository structure

Path	Contents

Path	Contents
apps/web/	React PWA: fan app, scorer app, admin console (route-split)
apps/api/app/engines/scoring/	Per-sport scoring from events + rule version
apps/api/app/engines/payouts/	Payout table generator and tie splitting (core IP: test first)
apps/api/app/engines/p2p/	Fill validation, quotes, settlement
apps/api/app/ledger/	Double-entry posting helpers
apps/api/app/jobs/	lock, settle, void, sweep
apps/api/tests/fixtures/	Worked examples from this PRD as golden tests
supabase/migrations/	Schema from section 7

10. Build roadmap

Phase 0 — Foundation (week 1)

- Repo, CI, Supabase project, auth with OTP, base PWA shell.

Phase 1 — Scoring, payout and P2P engines (weeks 2–3)

- Pure-Python engines for all three sports, payout generator and P2P settlement.
- Golden tests from every worked example in section 6. **No UI work starts until this suite passes.**

Phase 2 — Data model and admin setup (week 4)

- Migrations for section 7, tournament/fixture/squad import, line-up publishing.

Phase 3 — Scorer app (week 5)

- Event entry, offline queue, dual-scorer reconciliation, admin confirmation.

Phase 4 — Fan app and free pilot (weeks 6–7)

- Lobby, team builder (uniform pricing, 10 teams), free contests, live points, leaderboards, 0-stake "bragging rights" challenges.
- **Pilot at a campus sports week.** No money involved.
- Use the pilot's real matches to validate the draft basketball and volleyball point values, then lock them as a new scoring rule version before paid launch.

Phase 5 — KYC, wallet and paid contests (weeks 8–9)

- KYC, Paystack deposits and withdrawals, ledger, paid contests, responsible-gaming limits.
- Gated on: fantasy licence, school approval, payment-provider approval.

Phase 6 — P2P Fantasy H2H (week 10)

- Board, direct challenges, escrow, settlement.

Phase 6a — P2P Match-result bets (week 11, licence-gated)

- Home/Away, auto-opposite side, partial fills, draw-void. Built behind a feature flag and switched on only with a sports-betting licence.

Phase 6b — Private leagues, referrals, Predict & Win (week 12)

- Join codes, referral bonuses, free pick'em.

Phase 7 — More sports and schools (week 13+)

- Next sports from the "coming soon" tiles; multi-school onboarding.

10.1 Roadmap summary table

Phase	Weeks	Deliverable	Gate to start
0	1	Foundation	—
1	2–3	Engines + golden tests	—
2	4	Data model + admin	Phase 1 tests green
3	5	Scorer app	Phase 2
4	6–7	Fan app + free pilot	Phase 3 + school permission for pilot
5	8–9	Paid contests	Fantasy licence + school approval + PSP approval
6	10	P2P Fantasy H2H	Phase 5
6a	11	P2P Match-result bets	Sports-betting licence
6b	12	Private leagues, referrals, Predict & Win	Phase 5
7	13+	More sports and schools	Pilot metrics met

11. Licensing, payments and compliance in Nigeria

> This section is a planning aid, not legal advice. Every point below goes to counsel before Phase 5.

11.1 Two products, two licence questions

- **Fantasy contests and Fantasy H2H:** skill-based fantasy, typically licensed by a state gaming regulator. FantasyFi describes itself as LSLGA-licensed (Lagos State Lotteries and Gaming Authority).
- **Match-result bets:** wagering on a sporting outcome. This is sports betting, which is usually a separate and stricter licence category. Peer-to-peer (exchange-style) betting may need specific permission.
- **Jurisdiction:** following the Supreme Court's November 2024 decision on the National Lottery Act, gaming regulation outside the FCT is a state matter. The licence needed depends on the school's state and where users play from.

11.2 Age and identity

The whole platform is 18+, including free mode. Sign-up requires a declared date of birth showing 18+. Paid play additionally requires verified age via NIN/BVN, and a KYC date of birth under 18 closes the account.

11.3 School approval

A formal agreement with the school (sports council or Dean of Student Affairs) covering use of team and athlete names, scorer volunteers and the paid launch. This is the most likely point of failure, because athletes staking on their own matches (section 13) is exactly the kind of issue a school will raise.

11.4 Payments

Paystack and Flutterwave classify gaming as a restricted category and typically ask for a licence before onboarding. Wallet balances held for users may raise questions about holding customer funds. Ask counsel and the PSP whether a segregated or trust account is needed.

11.5 Responsible gaming

- Deposit and daily stake limits (default caps set by admin).
- Self-exclusion (24h, 7 days, 30 days, permanent).
- Visible "18+ only, play responsibly" messaging, as FantasyFi shows.

11.6 Data protection (NDPA 2023)

KYC data is sensitive personal data: encrypt at rest, minimise retention, publish a privacy policy, and register with or file to the NDPC as required.

11.7 Questions for counsel

1. Which state licence covers fantasy contests for a campus in our state, and what does it cost and take?
2. Is user-vs-user Fantasy H2H with a commission treated as fantasy or as betting?
3. Can Home/Away P2P bets (exchange-style, with partial fills) be licensed, and under which category?
4. Does holding user wallet balances require a segregated account or other licence?
5. Tax treatment of rake and commission revenue, and any withholding on winnings.
6. What does the school agreement need to cover (names, likeness, athlete participation)?

12. Go-to-market and marketing strategy

12.1 Core positioning

"Your campus, your league." The only fantasy game where you can pick the striker from your hall.

12.2 Launch at a sports week

Pilot free during the school's biggest tournament week, with sponsor prizes for the season leaderboard (Phase 4).

12.3 Campus ambassadors

One ambassador per faculty or hall, rewarded with referral bonuses once paid play opens.

12.4 Community channels

WhatsApp and Telegram channels per sport for line-up alerts and contest closing reminders. FantasyFi runs a Telegram community and a "buzz" ticker of lock countdowns.

12.5 Rivalries as a product feature

Faculty-versus-faculty and hall-versus-hall aggregate leaderboards and P2P challenges between rival fans.

12.6 Sponsor-guaranteed pools

Local businesses top up Guaranteed contests in exchange for branding on the contest card.

12.7 Metrics to track from day one

- Sign-ups per faculty, contest fill rate, P2P match rate, D7 retention, scorer turnaround, dispute rate.

13. Risks and open questions

Risk	Impact	Mitigation
Accepted risk: athletes may stake on their own matches, including against their own team	Match-fixing incentive; reputational damage; school may withdraw approval	Accepted by the product owner. Scorer dual-entry and admin sign-off still apply. Revisit before paid launch with the school.
40% platform fee is well above FantasyFi's ~10–30%	Prizes look small next to national apps; 1v1 contests return only 1.2× entry	Cards lead with prize pool and 1st prize; sponsor-Guaranteed pools; push 1v1 players to H2H challenges; fee is a config value
Licences take longer than the roadmap	Paid phases slip	Free pilot first; match-result bets behind a flag
Payment provider refuses onboarding	No deposits	Approach both Paystack and Flutterwave early with the licence plan
₦1,000 minimum entry is high for some students	Low fill rates	₦400 Mini ticket on every match; track fill rate by tier; minimums are config values
Scorer error or bias	Wrong payouts	Two scorers, reconciliation, 24h dispute hold, scorers barred from their matches
Uniform pricing causes many ties	Smaller prizes, less differentiation	Equal tie splits, C/VC/Key, 10 teams; dynamic pricing later
Collusion between linked accounts in P2P	Chip-dumping, laundering	Linked-account blocks, KYC-name withdrawals, repeat-pair flags
Low liquidity in large contests	Contests cancelled	Pools scale with entries, minimum fill, sponsor guarantees

13.1 Open questions to resolve during the build

1. Licence category and cost for each product, from counsel (section 11.7).

14. Appendix — instructions for building this with Claude Code

14.1 Suggested CLAUDE.md content for the repository root

```
# CampusXI — build context for Claude Code
This repo implements the CampusXI PRD (CampusXI_PRD.docx).
1. The scoring, payout and P2P engines (apps/api/app/engines/) are the core IP. Never change
them without running the golden tests built from PRD section 6 worked examples.
2. stat_events, p2p_bet_events, wallet_ledger and audit_log are append-only. Never UPDATE or
DELETE rows; corrections are new reversing rows.
3. All money is integer kobo. No floats anywhere in money paths.
4. Follow the schema in PRD section 7 and API contracts in section 8 unless told otherwise.
5. Stack: React+TS+Vite PWA, FastAPI, Supabase (Postgres, Auth, Realtime), APScheduler, Resend,
Paystack. Flag before adding anything else.
6. Build in the phase order of section 10. No UI before Phase 1 tests pass. Match-result bets
stay behind a feature flag.
```

14.2 Suggested first prompts to Claude Code, in order

1. "Set up the monorepo from PRD section 9.5 with FastAPI, a Vite React PWA and a Supabase project. Don't build features yet."
2. "Implement the football, basketball and volleyball scoring engine from section 6, with golden tests."
3. "Implement the payout engine from section 6.8 and make every table in section 6.9 a golden test."
4. "Implement P2P quotes and settlement from sections 6.11–6.12 with the worked example as a test."
5. "Create migrations for section 7, enforcing append-only tables with triggers."
6. "Build the scorer app (Epic E) with an offline queue."

14.3 A final note on scope discipline

Ship the free pilot first. Every paid feature depends on licences and approvals that the code can't speed up, and the pilot proves demand, scorer reliability and the right price points before any money moves.